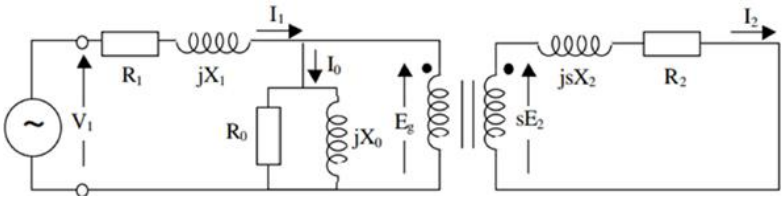
EEE303 Tutorial 4a

1. A 220-V, three-phase, two-pole, 50-Hz induction motor is running at a slip of 5%. Find
   (a) The speed of the magnetic fields in revolutions per minute
   (b) The speed of the rotor in revolutions per minute
   (c) The slip speed of the rotor
   (d) The rotor frequency in hertz
2. Answer the questions in Q1 for a 480-V, three-phase, four-pole, 60-Hz induction motor running at a slip of 3.5%.
3. A three-phase, 60-Hz induction motor runs at 890 r/min at no load and at 840 r/min at full load.
   (a) How many poles does this motor have?
   (b) What is the slip at rated load?
   (c) What is the speed at one-quarter of the rated load?
   (d) What is the rotor's electrical frequency at one-quarter of the rated load?
4. The parameters of an induction motor when the rotor is blocked are given as below:

   $R_1 = 0.15\ \Omega$ $X_1 = 0.38\ \Omega$ $R_2 = 0.11\ \Omega$ $X_2 = 0.39\ \Omega$
   $R_0 = 0.05\ \Omega$ $X_0 = 12.0\ \Omega$ $V_1 = 254\ \text{V}$ $a_{eff} = 1.1$

   For slip = 0.07, derive and calculate the rotor reactance X_2 and resistance R_2 as referred to the stator, then draw the equivalent circuit.
5. A 480 V_L, 50 Hz, 3 phase induction motor draws a 120 A at 0.87 PF lagging. The stator and rotor copper losses are 4.1 kW and 1.5 kW, respectively. While the friction and windage losses are 0.9 kW, the core losses are 2.1 kW, and the stray losses are 0.5% of the input power. Find
   (a) The input power P_{in} and air-gap power P_{AG}
   (b) The power converted from electrical to mechanical form P_{mech}
   (c) The output power P_{out}
   (d) The efficiency of the motor
6. A 20 hp, three-phase, 50 Hz, 4 pole induction motor has a full-load slip of 4 per cent. The friction and windage losses are 500 W. Calculate the rotor copper loss and rotor speed.